

- 39** A patient with kidney failure can be given a kidney transplant. In a kidney transplant, a healthy kidney is taken from another person and put into the patient's body.

Sometimes the donated kidney cells are killed by the patient's body. In this primary immune response, T-lymphocytes bind to antigens on the donated kidney cells and are activated.

The statements describe the next stages of the primary immune response.

- 1 Activated T-lymphocyte secretes cytokines.
- 2 B-lymphocytes differentiate into plasma cells and make antibodies.
- 3 T-killer cells bind to antigens on donor kidney cells.
- 4 T-killer cells release toxins that destroy the donor kidney cells.
- 5 T-lymphocytes multiply by mitosis and differentiate into T-killer cells.

Which order correctly explains how donor kidney cells are killed?

- A** 1 → 5 → 2 → 3
- B** 1 → 5 → 3 → 4
- C** 3 → 1 → 5 → 2
- D** 3 → 5 → 4 → 1